Matteo Guardamagna, **PhD**

Solberg Postdoctoral Fellow at *Kavli Institute for Systems Neuroscience* (NTNU)

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https://matteoguardamagna.github.io/

Research Interests

My research investigates the **developmental algorithms** by which neural circuits assemble into functional computational units and support the emergence of natural behaviors. By combining large-scale in vivo electrophysiology in developing rodents with topological data analysis and quantitative 3D behavioral decomposition, I chart how thalamocortical networks self-organize, calibrate their dynamics using the body as a training ground, and ultimately bind to the external world. My long-term goal is to uncover the shared developmental primitives of thalamocortical computations across **sensory**, **motor**, and **cognitive systems**.

Education

2018-2022:	**PhD** in **Science**, Radboud University (Donders Institute), Nijmegen (NL). Thesis: Temporal Coding Dynamics in the CA1 region of the hippocampus Graduated Cum Laude Distinction
2015-2017:	**MSc** in **Cognitive Neuroscience**, University of Padua, Padua (IT). Thesis: Functional characterization of spatially-modulated neurons in the dorsal subiculum of the mouse. Graduated cum laude (110/110 cum laude)
2012-2015:	**BSc** in **Psychological Sciences**, University of Milano-Bicocca, Milan (IT). Thesis: Cross-modal influences on face perception: psychophysiological study Graduated cum laude (110/110 cum laude)

Research Positions

2023-present:	**Postdoctoral Fellow** Kavli Institute for Systems Neuroscience (NTNU), Trondheim (NO) Supervisors: Prof. Edvard Moser and Prof. May-Britt Moser
2018-2022:	**PhD Candidate** (Marie-Curie Early-Stage Researcher) Donders Institute (Radboud University), Nijmegen (NL). Supervisors: Prof. Francesco Battaglia and Dr. Federico Stella
2020-2021:	**Visiting PhD Student** (5 Months) Kavli Institute for Systems Neuroscience (NTNU), Trondheim (NO) Supervisors: Prof. Clifford Kentros
2016-2017:	**Visiting Research Student** (MSc Thesis) University College London (UCL), London (UK) Supervisors: Prof. Francesca Cacucci and Prof Tom Wills
2014-2015:	**Undergraduate Research Student** (BSc Thesis) University of Milano-Bicocca, Milan (IT). Supervisors: Prof. Alice Mado Proverbio

Grants and Awards

2025-present	**Solberg** Postdoctoral Fellowships (Kavli Institute; 80000€ per year)
2023-2025:	**EMBO** Postdoctoral Fellowships (ALTF 738-2022; total 148000€)
2018-2021:	**Marie Skłodowska-Curie** Fellowship for Early Stage Researchers (total 374000€)
2016-2017:	**MIUR** Scholarship for Excellent Traineeships (total 500€)
2016-2017:	**Erasmus+** Traineeship Scholarship (total 3000€)

2022	**Cum Laude Distinction** from the PhD in Science at Radboud University.
2017	Graduated **cum laude** (110/110 cum laude) from the M.Sc.
2015	Graduated **cum laude** (110/110 cum laude) from the BSc.

Publications

- Chadney*, O. M. T., **Guardamagna*, M**., Dörr, F., Descamps, L. A. L., Stella, F., Battaglia, F., & Kentros, C. G. (2026). The emergence of a novel CA1 spatial map requires direct entorhinal input. bioRxiv. https://doi.org/10.64898/2026.07.02.736055

- **Guardamagna, M**., Hermansen, E., Carpenter, J., Lykken, C. M., Dunn, B. A., Moser, E. I., & Moser, M. (2026). Toroidal topology of grid-cell activity precedes spatial navigation during development. bioRxiv (Cold Spring Harbor Laboratory). https://doi.org/10.64898/2026.03.10.710908

- **Guardamagna*, M.**, Chadney*, O., Stella*, F., Zhang, Q., Kentros, C., & Battaglia, F. P. (2025). Direct entorhinal control of CA1 temporal coding. Nature Communications, 16(1), 6430. https://doi.org/10.1038/s41467-025-61453-2

- Lykken, C. M., Kanter, B. R., Nagelhus, A., Carpenter, J., **Guardamagna, M.**, Moser, E. I., & Moser, M. (2025). Functional independence of entorhinal grid cell modules enables remapping in hippocampal place cells. bioRxiv (Cold Spring Harbor Laboratory). https://doi.org/10.1101/2025.09.24.677985

- Modi, B., **Guardamagna, M.**, Stella, F., Griguoli, M., Cherubini, E., Battaglia, FP (2023). State Dependent Coupling of Hippocampal Oscillations. eLife, https://doi.org/10.7554/eLife.80263.

- **Guardamagna**, **M.**, Stella, F., & Battaglia, F. P. (2023). Heterogeneity of network and coding states in mouse CA1 place cells. Cell Reports, 42(2), 112022. https://doi.org/10.1016/j.celrep.2023.112022

- **Guardamagna, M.**, Eichler, R., Pedrosa, R., Aarts, A., Meyer, A., & Battaglia, F. (2022). The Hybrid Drive: A Chronic Implant Device Combining Tetrode Arrays with Silicon Probes for Layer-Resolved Ensemble Electrophysiology in Freely Moving Mice. J. Neural Eng. 19 036030.

- Proverbio, C. A. A. M., Nasi, V. L., Arcari, L. A., De Benedetto, F., **Guardamagna, M.**, Gazzola, M., & Zani, A. (2015). The effect of background music on episodic memory and autonomic responses: Listening to emotionally touching music enhances facial memory capacity. Scientific Reports, 5. https://doi.org/10.1038/srep15219

- Proverbio, A. M., Manfrin, L., Arcari, L. A., De Benedetto, F., Gazzola, M., **Guardamagna, M.**, … Zani, A. (2015). Non-expert listeners show decreased heart rate and increased blood pressure (fear bradycardia) in response to atonal music. Frontiers in Psychology, 6(OCT). https://doi.org/10.3389/fpsyg.2015.01646

Invited Talks and Poster Presentations

Invited talks:

- 26.11.2026. **SWC/UCL** (UK). *On the origin of our sense of space: toroidal manifolds in grid-cell networks emerge before navigational experience* (ENNS).
- 15.10.2026. **ICM Paris** (FR). *On the origin of our sense of space: the emergence of spatial networks before active navigation.*
- 07.10.2025. **Columbia University** (USA). *Ontogeny of Toroidal Manifolds in the entorhinal cortex.*
- 06.10.2025. **Johns Hopkins University** (USA). *Ontogeny of Toroidal Manifolds in the entorhinal cortex* (Kavli NDI Series)
- 24.05.2025. **Spring Hippoccampal Research Conference** (IT). *Ontogeny of the grid network.*
- 11.12.2023. **Donders Institute** (NL). *Development of Toroidal Topology in the Grid Cells Network* (Memory Development in Mammalian Species Symposia).
- 30.5.2023. **Spring Hippocampal Research Conference** (IT). *Direct Entorhinal Control of CA1 Temporal Coding. Spring Hippocampal Research Conference. Verona* (IT).
- 14.4.2023. **UCL** (UK)**.** Heterogeneity of Network and coding states in mouse CA1 place cells. Journal Club, Wills-Cacucci – UCL. London (UK). (Virtual).
- 28.4.2021. **INMED** (FR). Medium gamma control of CA1 place cells' phase precession. Journal Club – Physionet Team (Virtual).

Posters presentations:

SfN 2026, FENS 2026, Neural basis of natural behaviors 2026; SfN 2025; Ascona Lake Conference 2024; SfN 2024; FENS 2024; SfN 2023; FENS 2022; SfN 2021; Donders Session 2020.

Mentoring, Teaching and other activities

2023-present: Supervising PhD student Jordan Carpenter (Moser Lab)
2020: Supervising MSc student and MSc Thesis of George Bardanikas (Battaglia Lab)
2019: Supervising MSc student and MSc Thesis of Kyrah Wilmink (Battaglia Lab)

2025-2026: **Lecturer on** "Neural Dynamics" for the NEVR3003,
2024-2025: **Lecturer on** "Neural Dynamics" for the NEVR3003,
2023-2024: **Lecturer on** "Neural Dynamics" for the NEVR3003,
Behavioral and Cognitive Neuroscience Course.
2020: **Teaching assistant** for Coding with Matlab (BSc Course).
2020: **Teaching assistant** for Cognitive Neuroimaging (MSc Course).
2019: **Teaching assistant** for Behavioral Neuroscience (BSc Course).

2026-present: **YROCK** committee member

Ad-hoc reviewer: Nature Communications, Journal of Neuroscience, Journal of Neuroscience Methods, Journal of Neural Engineering, Cell Reports, Progress in Neurobiology, Neuron (with Prof. F.Battaglia), eLife (with Prof. F.Battaglia).